

Hamza Akram

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Professional Summary

Results-driven Software Engineering postgraduate with practical experience designing, developing, testing, and maintaining software applications. Proficient in Python, SQL, software debugging, database management, and AI-assisted development, with a strong foundation in object-oriented programming, data structures, algorithms, and software engineering principles. Currently working remotely as a Generalist Expert with Mercor, evaluating large language model outputs for factual grounding, relevance, and reasoning quality against source material. Independently designed and built two privacy-first applications - Aura and TruthLens - showcasing end-to-end ownership from system architecture through deployment. Passionate about technology, continuous learning, and using innovative solutions to solve real-world challenges. Currently pursuing an MSc in Software Engineering at the University of Hertfordshire and eligible to work in the UK.

Skills

- **Programming Languages:** Python, JavaScript, TypeScript, SQL, HTML5, CSS3
- **Artificial Intelligence & Applied AI:** Large Language Models (LLMs), AI Model Output Evaluation, Prompt Engineering, Local LLM Integration (Ollama)
- **Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Design Patterns
- **Web Development:** React.js, Node.js, FastAPI, RESTful API Development, Frontend & Backend Development, Third-Party API Integration
- **Cloud & Tools:** AWS (EC2, Lambda, S3, SQS, SNS, CloudWatch, IAM, VPC), Git, GitHub, Visual Studio Code, Postman
- **Networking:** TCP/IP, DNS, HTTP/HTTPS, Routing, Firewalls, Load Balancing
- **Systems Administration:** Linux and Windows Server Administration
- **Technical Support & Communication:** Diagnosing complex technical issues and communicating solutions clearly to stakeholders of varying technical levels
- **Soft Skills:** Problem Solving, Analytical Thinking, Technical Communication, Team Collaboration, Customer-Focused Development, Adaptability, Continuous Learning

Projects

Aura - Offline, Privacy-First Personal AI Assistant github.com/hamzaakrm/aura

- Built a fully offline desktop AI assistant in Python that runs entirely on-device, ensuring no user data ever leaves the machine - applying strong data-privacy and security-by-design principles.
- Integrated local LLMs via Ollama for offline conversational AI, with persistent long-term memory, automatic summarization, and multilingual offline voice input/output (speech-to-text and text-to-speech).
- Designed a cross-platform desktop GUI (PySide6) backed by SQLite for local storage, with computer-automation features and an optional "Learning Mode" for web research and content tasks.

TruthLens - AI-Powered Fact-Checking & Source Verification Tool github.com/hamzaakrm/TruthLens

- Developed a privacy-focused research tool (FastAPI backend, local LLM via Ollama) that searches the web, cross-references multiple sources, and generates cited answers while explicitly flagging uncertainty and source disagreement.
- Implemented duplicate/wire-story detection via similarity clustering, automated fact-vs-opinion classification, and Wayback Machine archive checks to flag sources edited after publication.
- Built local source-reputation tracking (SQLite), image verification (EXIF metadata and AI-generation signal detection), and exportable Markdown/PDF reports with multilingual cross-referencing.

Work Experience

Generalist Expert, AI Model Evaluation - Mercor (Remote, Contract)

5 August 2026 – Present

- Evaluate large language model outputs against source material, assessing claim by claim whether a statement is substantiated or only sounds precise.

- Identify unsupported causal attributions and fluent filler that reads as insight but adds no information beyond the given context.
- Judge relevance in context, where identical phrasing can be substantive in one position of a document and redundant in another.
- Apply grading criteria consistently across a high volume of independent judgments, including borderline cases that resist an easy call.
- Produce structured written justifications for each judgment to support reviewer consistency and downstream data quality.

Developer - Triolinks Services (Pakistan)

March 2024 – June 2025

- Developed and maintained Python-based applications to support business and healthcare management processes.
- Designed and implemented backend functionalities using Python, improving application performance and reliability.
- Worked with SQL databases to store, retrieve, and manage application data efficiently.
- Assisted in integrating AI-powered features to enhance automation and user experience.
- Participated in software testing, debugging, and troubleshooting to ensure high-quality deliverables.
- Collaborated with senior developers throughout the Software Development Life Cycle (SDLC), including requirements gathering, development, testing, and deployment.
- Utilized Git for version control and team collaboration.
- Prepared technical documentation and contributed to project planning discussions.

Education

University of Hertfordshire

MSc Software Engineering

Sept 2025 – Sept 2027

- Measure and Models for Software Engineering (Software Metrics, UML Modelling, System Analysis and Design, Requirements Engineering, Process Modelling, Quality Assurance, Software Performance Evaluation, Risk Analysis, and Software Process Improvement)
- Software Engineering Practice and Experience (Agile Development, Team Collaboration, Software Development Life Cycle (SDLC), Version Control using Git, Software Testing, Code Reviews, Requirements Analysis, Project Delivery, Problem Solving, and Industry Best Practices)
- Programming for Software Engineers
- Team Research and Development Project (Group-based Software Development Project, Collaborative Research, Agile Methodology, Requirement Analysis, System Design, Implementation and Testing, Version Control using Git, Team Communication, Problem Solving, and Final Project Delivery/Presentation)
- Responsible Technology (Legal, Social, Ethical and Professional Issues)

University of Haripur

BSc Software Engineering (Hons)

February 2020 – February 2024

- Object-Oriented Programming (Java, Python, OOP Principles, Design Patterns, Application Development)
- Software Architecture and Design (System Design, Architectural Patterns, UML Modelling, Scalable Software Solutions)
- Database Systems (SQL, Database Design, Data Modelling, Query Optimization, Database Management)
- Web Development Technologies (HTML, CSS, JavaScript, Front-End and Back-End Development)
- Artificial Intelligence and Machine Learning (Python, Anaconda, Jupyter Notebook, Algorithms)
- Cloud Computing (Cloud Architecture, AWS Services, Virtualization, Cloud Deployment and Management)
- Cyber Security Fundamentals (Network Security, Risk Assessment, Security Controls, Data Protection)
- Data Structures and Algorithms (Algorithm Design, Problem Solving, Data Management)
- Software Project Management (Risk Management, Agile Project Delivery, Team Collaboration)
- Research Methods for Computing (Agile Project Delivery, Team Collaboration)
- Operating System (Linux, Windows, Memory Management, System Administration)